

APUKA, Russian Federation

WMO: 259560

Lat: 60.443N

Lon: 169.608E

Elev: 15

StdP: 14.69

Time Zone: 12.00 (E12)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-15.9	-12.3	-26.0	1.3	-14.7	-22.3	1.6	-10.8	45.8	18.4	42.8	16.2	10.1	50	0.854

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	7.5	63.1	56.8	60.5	55.2	58.6	54.2	57.7	61.5	56.1	59.5	54.7	57.8	10.0	270

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
55.6	65.8	59.9	54.0	62.0	58.0	52.8	59.3	56.6	25.0	61.7	23.9	59.9	23.0	58.0	63.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 36.9	31.8	27.4	DB	-22.5	69.2	6.9	3.2	-27.4	71.5	-31.4	73.3	-35.2	75.1	-40.2	77.4	(4)
(5)			WB	-22.9	61.5	6.7	1.7	-27.7	62.7	-31.7	63.7	-35.5	64.7	-40.4	65.9	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	29.8	8.7	9.4	13.6	23.8	35.9	45.6	53.0	53.5	46.6	33.1	21.4	12.1
	DBStd	18.48	13.17	11.61	11.68	7.72	3.89	3.73	4.04	3.12	4.54	7.38	10.38	11.55
	HDD50	7602	1281	1137	1129	786	436	139	16	6	115	524	858	1175
	HDD65	12833	1746	1557	1594	1236	901	582	372	357	551	989	1308	1640
	CDD50	244	0	0	0	0	0	7	109	114	14	0	0	0
	CDD65	0	0	0	0	0	0	0	0	0	0	0	0	0
	CDH74	0	0	0	0	0	0	0	0	0	0	0	0	0
	CDH80	0	0	0	0	0	0	0	0	0	0	0	0	0
Wind	WSAvg	11.3	15.9	15.2	14.2	10.6	7.4	7.2	7.2	7.5	8.8	11.2	14.2	16.0
Precipitation	PrecAvg	18.3	1.1	0.9	1.3	0.8	0.7	1.4	1.3	2.6	2.4	2.4	2.0	1.3
	PrecMax	27.1	3.6	2.6	2.8	1.8	2.8	5.3	4.9	6.2	5.4	5.6	4.3	3.1
	PrecMin	11.8	0.0	0.0	0.0	0.3	0.2	0.3	0.1	0.0	0.4	0.0	0.4	0.0
	PrecStd	3.7	1.0	0.6	0.9	0.4	0.6	1.1	1.0	1.4	1.3	1.5	1.2	1.1
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	36.2	36.3	35.8	41.0	51.0	60.9	68.0	65.6	60.2	50.3	40.5	36.6
		MCWB	34.1	33.7	33.5	37.7	45.7	52.3	59.0	58.4	54.3	45.1	38.9	34.7
	2%	DB	34.4	32.5	33.4	37.9	46.9	55.7	64.6	62.2	56.9	47.2	38.5	34.6
		MCWB	32.2	30.3	31.5	34.6	43.1	51.3	57.9	56.2	51.7	43.4	37.2	32.8
	5%	DB	31.9	30.0	31.9	35.7	44.2	53.2	61.5	60.1	54.8	45.0	36.6	32.7
		MCWB	29.9	28.4	30.4	32.8	41.2	49.3	56.1	55.0	50.6	41.8	34.8	30.9
	10%	DB	27.4	26.9	30.2	34.1	41.9	51.1	59.3	58.5	53.2	43.0	34.8	29.4
		MCWB	25.5	25.0	28.7	31.6	38.9	47.6	55.0	54.1	49.6	39.7	32.9	27.6
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	34.4	34.1	34.0	38.1	46.3	54.4	61.1	59.4	55.5	46.1	39.3	35.1
		MCDB	35.8	36.3	35.3	40.6	50.5	58.6	66.1	64.7	59.2	49.3	39.9	36.2
	2%	WB	32.5	30.8	32.1	34.9	43.7	51.5	58.8	57.0	52.9	43.9	37.2	32.9
		MCDB	34.1	32.2	32.9	37.2	46.8	55.0	62.9	60.7	55.7	46.7	38.1	34.3
	5%	WB	29.7	28.2	30.6	33.2	41.3	49.7	57.0	55.7	51.5	42.2	35.0	31.0
		MCDB	31.7	29.7	31.8	35.2	44.0	52.8	60.4	59.2	54.0	44.9	36.2	32.7
	10%	WB	25.6	25.3	28.6	31.9	39.1	48.2	55.5	54.5	50.2	40.2	33.2	27.7
		MCDB	27.2	27.0	30.1	34.0	41.5	50.6	58.6	57.7	52.7	42.8	34.7	29.4
Mean Daily Temperature Range	5% DB	MDBR	9.3	10.4	12.3	11.0	7.5	6.9	7.6	7.5	8.9	8.9	8.5	9.3
		MCDBR	7.8	12.1	8.7	9.4	10.8	11.2	12.1	11.1	11.4	8.9	7.8	8.0
		MCWBR	7.6	11.7	8.4	8.1	8.5	8.5	8.2	8.3	9.1	7.5	7.6	8.6
	5% WB	MCDBR	8.0	11.0	8.2	8.1	10.9	10.2	10.5	9.5	9.5	8.5	7.6	8.7
		MCWBR	8.1	11.1	8.2	7.4	8.6	8.0	8.1	7.6	8.1	7.5	7.6	9.4
Clear-Sky Solar Irradiance	taub	0.236	0.284	0.336	0.383	0.434	0.406	0.435	0.428	0.340	0.314	0.266	0.218	
	taud	2.077	2.130	2.040	1.963	1.986	2.252	2.206	2.194	2.456	2.314	2.106	1.988	
	Ebn at Noon	165	221	243	252	248	261	248	239	249	210	146	116	
	Edh at Noon	13	23	36	47	50	39	40	37	24	20	13	9	
All-Sky Solar Radiation	RadAvg	116	356	837	1425	1586	1569	1575	1158	800	406	130	51	
	RadStd	18	48	127	119	190	184	139	134	96	44	24	7	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (1 neighbor)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page